

AI-Driven Polypharmacy Risk Prediction Using Graph Neural Networks

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Abstract—Polypharmacy, which is the usage of several drugs simultaneously, is becoming a common phenomenon, especially among elderly patients and those with long-term illnesses. Although it may be essential to the successful treatment, polypharmacy is a leading contributor to the occurrence of adverse drug reactions (ADRs), drug-drug interactions (DDIs), medication errors, and decreased therapeutic efficacy. The classical rule-based clinical decision support systems are primarily concerned with the associated pairwise drug interactions and are based on a predefined base of medical knowledge. But in treatment settings where a combination of drugs is available, these methods tend to fail to represent the complicated dynamics that arise when multiple drugs interact at the same time. In this project, we present a proposal of an artificial intelligence-based system for predicting polypharmacy risk based on Graph Neural Networks (GNNs). The model proposed is that every drug is a node of a graph, and drug-drug interactions and pharmacological relationships are edges between the nodes. This multi-drug interaction is an effective way in which this graph-based representation enables the system to model the complex interaction patterns between drugs. The model is trained by using deep learning algorithms on graph-structured biomedical data to make predictions that are important representations of drug interactions with the probability of an adverse reaction developing upon medication combinations. The given system combines drug interaction data sets and takes advantage of the possibility of GNNs to spread information between interconnected nodes and identify known and never-seen interaction patterns. The experimental assessments prove the given method to be more accurate in making predictions, generalizing, and being more robust in comparison with the classical machine learning and rule-based methods.

Keywords: Polypharmacy, Graph Neural Networks, Drug-Drug Interactions, Adverse Drug Reactions, Artificial Intelligence in Healthcare, Clinical Decision Support Systems, Deep Learning, Biomedical Data Analysis

I. INTRODUCTION

Polypharmacy, which is characterized by the co-administration of several distinct medications by a person, has gained prominence in contemporary healthcare systems. It is also highly common in geriatric patients and those with chronic conditions like diabetes, cardiovascular diseases, and hypertension. Although taking several medications may be required to help treat complicated health conditions, there are a number of significant risks associated with taking more than one medication: increasing the risk of adverse drug reactions (ADRs), drug-drug interactions (DDIs), medication errors, and decreasing the effectiveness of treatment. These risks are very difficult for healthcare providers, and they can cause a higher rate of hospitalization, an elevation of medical expenses, and jeopardization of patient safety [1] [2].

Conventional clinical decision support systems used for drug interaction detection are mainly rule-based and rely on a preset body of medical knowledge. These systems largely focus on the detection of pairwise drug interactions, when the interaction between two drugs is determined separately. In clinical practice, however, patients have many medications at once, and the multi-drug interaction patterns are very complicated and may not be adequately represented by simple pairwise analysis [3]. This means that the traditional approaches might not see the non-obvious or indirect correlations between drugs, and they would not be good at predicting the potential adverse effects.

Recent progress in machine learning and artificial intelligence has created new prospects in the large-scale analysis of biomedical data and unravelling intricate drug interaction patterns. Graph Neural Networks (GNNs) have become one of the potent methods of relationship modeling in graph-structured data [4] [5]. When considering drug interaction analysis, drugs

may be represented as nodes in a graph, and any drug-drug interaction may be represented as an edge between nodes. This representation allows modeling the intricate relationships and learning meaningful patterns of interrelated biomedical data by the system [6].

The present paper suggests an AI-based polypharmacy risk prediction model using graph neural networks. The given solution creates a drug interaction graph according to which deep learning methods are used to learn the latent interactions between a set of drugs [7] [8]. With the help of graph-based learning, the system is able to estimate known and unknown drug interactions that cause adverse reactions with the use of complex drug combinations [9] [10].

The given model is expected to become more effective in terms of clinical decision support systems, offering early warning of harmful drug combinations. The system can help health practitioners make better and safer treatment choices by enhancing the quality of predicted drug interactions. Finally, this study will enhance patient safety, decrease the negative drug events, and contribute to the use of artificial intelligence in healthcare analytics.

II. LITERATURE SURVEY

Polypharmacy can be defined as the use of a combination of more than one drug at a time to cure the complex diseases like diabetes, cardiovascular disorders, and cancers. Although it will enhance the outcomes of treatment, the approach will increase the risk of drug-drug interactions (DDIs) and adverse drug reactions remarkably. Conventional clinical trials and laboratory experiments to determine such interactions are costly and time-consuming. Consequently, scholars have examined Artificial Intelligence (AI) and Graph Neural Networks (GNNs) to forecast the risks of polypharmacy in a more effective way [11] [12].

The Decagon model suggested by Zitnik et al. is one of the most significant publications in this field, where Graph Convolutional Networks (GCN) are applied to predict polypharmacy side effects. The model builds a multimodal graph that comprises drug-protein interactions, protein-protein interactions, and drug-drug interactions [13] [14].

Every side effect is associated with a distinct type of edge in the graph, which enables the model to acquire complex associations between medications. The method has been shown to greatly increase the accuracy of prediction relative to the traditional machine learning algorithms, and the use of graphs has proven effective in learning biomedical data.

Subsequent research was aimed at enhancing interpretability and prediction. As an example, the Graph Feature Attention Network (GFAN) was proposed to forecast side effects of polypharmacy by attaching varying attention weights to the target genes related to the combination of drugs. Such an approach can be used to determine the biological factors that contribute most to the predicted side effects, making it more interpretable [15]. The experimental findings demonstrated

competitive performance as compared to the past graph attention models and gave more precise explanations of predictions. [16] [17]

Recent studies have investigated more sophisticated GNN designs to predict drug interactions and risk levels. As an example, the Relational Graph Convolutional Network with multi-head attention (AERGCN-DDI) was suggested to enable the classification of the risk level of drug combinations rather than predicting the existence of an interaction. The method combines relational learning and attention processes to learn intricate associations between drugs and enhance the accuracy of prediction.

Other researchers have explored other GNN models to enhance the prediction of DDI, including Graph Attention Networks (GAT), GraphSAGE, and Graph Diffusion Networks. These models acquire drug representations according to their molecular structure and interactions. The experimental analyses based on several datasets have shown that GNN-related models are more effective compared to traditional machine learning models in detecting possible adverse drug reactions and polypharmacy.

Also, more complex systems, including multi-scale graph neural processes with cross-drug co-attention, have been proposed to learn hierarchical structural data of drug molecules and to deliver uncertainty-aware predictions. Such models integrate the multi-scale message passing and attention to increase prediction accuracy and reliability in the context of pharmacovigilance [3].

In general, the literature shows that Graph Neural Networks are very useful in complex biomedical relationships modeling, particularly drug interactions and polypharmacy risks forecasting. GNNs are able to identify more complex biological connections not present in the traditional models because they model drugs, proteins, and side effects as nodes and edges in a graph [7] [1]. Nevertheless, issues such as data scarcity, model interpretability, and clinical validation are still under research. Future studies can be aimed at combining electronic health records, genomic data, and explainable AI methods to create more accurate and individualized systems of polypharmacy risk prediction.

III. OBJECTIVES

A. Polypharmacy Risk Prediction by Artificial Intelligence

The ultimate goal of the current piece of work is to devise and produce an AI-based framework that could predict the possible risks related to polypharmacy based on the state-of-the-art deep learning algorithms. Polypharmacy (meaning simultaneous use of multiple medications) is typical of patient groups with chronic ailments and the geriatric population.

Nonetheless, it is highly dangerous as it can dramatically contribute to the risk of adverse drug reactions and dangerous drug-drug interactions. The proposed system provides a graph-based approach to learning in terms of interconnected networks between drugs, patients, and diseases by using graph-based approaches like Graph Neural Networks. The latter enables the system to identify complex medication

to medication dependency and accurately identify high-risk medication combinations. The aim is to enhance the precision of predictions and help medical workers make more secure prescriptions.

B. Network Modeling Drug-Drug Interaction Networks

The other notable goal is to model drug interactions and biomedical associations in the form of graph structures. Here, drugs, proteins, and diseases are depicted as nodes, and their interactions are depicted as edges. Graph-based modeling allows the system to examine indirect relationships and concealed patterns that are not easily revealed by the more traditional statistical or machine learning methods. With the use of the graph-based message passing methods in the context of Graph Neural Networks, the system is taught to acquire meaningful representations of drugs and the patterns of their interactions. This goal will assist in the identification of unreported or unreported drug interaction risks and enhance the general predictability of the prediction model.

C. In-time Identification of Adverse Drug reactions

Another goal of the system is to provide an opportunity to identify the possible adverse drug reactions in the case of complex drug interactions early. Anticipation of these reactions during the prescription of drugs can help avoid complications in the healthcare of patients to a large extent and minimize patient injuries. The suggested AI model examines the large-scale biomedical data, such as databases of drug interactions and patient medical histories, to identify trends related to adverse side effects. Early detection of risk will enable healthcare providers to change the course of treatment, alter the doses, or choose safer alternative medications. This goal eventually leads to positive healthcare management and patient safety.

D. Healthcare Professional Clinical Decision Support

The other crucial goal is to improve clinical decision support frameworks through the adoption of AI-based risk prediction frameworks in health care settings. The system is able to give predictive information to doctors, pharmacists, and medical researchers about the potential drug interactions and side effects. The insights will help clinicians in assessing the safety of complicated drug regimens and choosing the best treatment schedules. The proposed framework can ensure evidence-based medical decision-making and minimize the risk of prescription errors because it provides evidence-based recommendations.

E. Scalable and Data-driven Healthcare Analytics

The last goal is to design an efficient and scalable platform that will be able to handle big biomedical data. Drug, patient, and clinical data produced by healthcare systems are in large quantities, and meaningful analysis of these huge volumes of data can only be achieved through advanced computational models. The suggested system applies AI methods to these datasets to study them effectively, recognize the high-risk

patterns of medications, and provide actionable insights. This goal assists in the creation of smart healthcare analytics tools that enhance treatment outcomes, decrease the costs of healthcare, and enhance safer medication in the contemporary medical system.

IV. METHODOLOGY

Methodology

The proposed system involves an artificial intelligence-based method with the use of Graph Neural Networks (GNNs) to estimate polypharmacy risks and determine possible drug interactions between patients taking several drugs. The methodology starts with the collection of drug interaction datasets and patient medication datasets, where the information about drug names, the severity levels, and the description of the potential side effects is provided. These data form the main input for constructing the predictive model. The data is preprocessed before it is trained into the model to eliminate inconsistencies and give it a consistent format. Names of drugs are coded and converted into number representations by way of a label encoding method to enable the machine learning model to handle them.

Following a preprocessing, the information about the interactions of drugs will be demonstrated in the graph structure. In this graph, the drugs are considered as nodes, and the relationship between two drugs is represented by an edge between the nodes. This representation enables the system to provide the complex relationships that exist between various medications in polypharmacy situations. The node features are represented with the help of an identity matrix and are used to tell the model a unique drug in the graph. The resulting graph data is input into the Graph Neural Network.

The proposed model is trained with the help of the Graph Convolutional Network (GCN) architecture, which is composed of numerous convolution layers. The layers allow the model to internalize information of immediate neighbors and learn to bring forth latent patterns in drug interaction networks. The learned node embeddings are then pooled together with pairs of drugs, and a fully connected layer is then employed to predict the likelihood of interaction between those drug pairs.

The model is trained in the stage of learning based on the set of known drug interaction pairs. The training algorithm is composed of the Adam optimization algorithm and a binary cross-entropy loss function to reduce the difference between the prediction and the actual result and enhance the performance of the model. To guarantee stable learning and convergence, the model is trained over a number of epochs. The performance of the system after training is determined by the prediction accuracy of the model by comparing the labels of the predicted and actual interactions.

The model is analysed by processing the patient medication records after it has been trained. The list of prescribed drugs is processed, and all possible combinations of drugs are generated, which in each case involves every patient. The trained GNN model is then predicted to provide the probability of interaction between each drug pair. When the probability

TABLE I: Comparison of Existing Works on Drug–Drug Interaction and Polypharmacy Risk Prediction

Ref No	Year	Method / Model	Dataset Used	Feature Selection Techniques	Accuracy
[1]	2018	Network Medicine Framework	Drug Interaction Network	Graph Feature Learning	0.94
[2]	2016	Graph Convolutional Network (GCN)	Graph Structured Dataset	Spectral Graph Features	0.93
[3]	2021	Artificial Intelligence Model	Clinical Polypharmacy Dataset	Patient Medical Features	0.90
[4]	2021	Deep Graph Learning	Drug Interaction Dataset	Graph Representation Learning	0.92
[5]	2020	Deep Learning Model	Drug–Target Interaction Dataset	Molecular Feature Extraction	0.91
[6]	2022	Graph Neural Network Survey Model	Biomedical Graph Dataset	Graph Embedding Techniques	0.93
[7]	2022	Graph Neural Network Model	Drug Discovery Dataset	Biomedical Graph Features	0.92
[8]	2019	Deep Neural Network (DNN)	Drug Interaction Dataset	Neural Feature Learning	0.90
[9]	2019	Graph Convolutional Network	Drug Interaction Network	Graph Structural Features	0.91
[10]	2018	Deep Learning Model	Drug Interaction Dataset	Drug Interaction Features	0.92
[11]	2018	Graph Attention Network (GAT)	Graph Dataset	Attention-based Features	0.94
[12]	2021	Graph Neural Network	Drug Interaction Dataset	Graph Representation Features	0.92
[13]	2017	GraphSAGE Model	Large Graph Dataset	Node Embedding Features	0.91
[14]	2019	GAMENet Model	Medication Recommendation Dataset	Memory Network Features	0.93
[15]	2017	Semi-supervised GCN	Graph Dataset	Graph Convolution Features	0.92
[16]	2019	Deep Learning Model	Biomedical Literature Dataset	Text Feature Extraction	0.89
[17]	2018	Graph Convolutional Network	Polypharmacy Side Effect Dataset	Drug Interaction Features	0.94

that is predicted is greater than a given preset value, the system regards it as a potential drug interaction. The system retrieves the relevant level of severity and potential side effects in the interaction database.

Lastly, the proposed outcomes are shown, and an interactive interface is created with the help of Streamlit. The interface enables users to add drug interaction datasets and patient medication datasets, see the model accuracy, and receive the possible interactions between patient medications. The output shows the interacting drug pairs clearly, the level of interaction probability scores, the level of severity, and potential side effects that can occur, which become more helpful in healthcare professionals making safer decisions concerning medication and minimize the risks related to polypharmacy.

V. IMPLEMENTATION

A. Programming Environment

The Python programming language is used to implement the proposed system; this programming language is popular in the application of machine learning and artificial intelligence. The Python language has very good libraries that can be used to process data, develop models and evaluate performance. It is implemented in development platforms like Jupyter notebook or Google Colab that are effective in conducting experiments and visualizing the findings.

B. Handling and Preprocessing of Data

The libraries utilized to process the drug interaction datasets obtained will be NumPy and Pandas. These libraries are useful in the organization and cleaning of the data such as eliminating duplicate records, working with missing values and transforming raw data into a structured format. Additional preprocessing involves drug identifier encoding, feature derivation, and data preparation to be used in graph modeling.

C. Graph Construction

During this phase, the data on drug interaction is converted into a form of a graph. The drugs are denoted by the nodes, and the drug-drug interaction is denoted by the edges. The graph

model is useful in modeling intricate interrelations between various drugs that are used in polypharmacy. The network provides the model with the ability to know the interaction of various drugs with one another.

D. Graph Neural Networks Model Development

The GNN techniques are used to formulate the predictive model. The neural network structure is developed with deep learning models like PyTorch or TensorFlow. The GNN model is trained on the graph structure, whereby the data gets transmitted among the networked nodes, and meaningful representations of drugs and their interactions are learned.

E. Model Training

This dataset will be separated into a training and testing set to train and test the model successfully. The Graph Neural Network is trained by acquiring patterns of known drug-drug interactions. Gradient descent is an optimization algorithm that is applied to adjust the model parameters and enhance the accuracy of prediction.

F. Prediction and Risk Analysis

The model is then used to predict possible risks of new drug combinations after training. The system examines various drugs consumed by a patient and reveals potential negative interactions that can be encountered because of polypharmacy.

G. Performance Evaluation

The evaluations of the implemented model are conducted on the basis of standard metrics of evaluation that include accuracy, precision, recall, and F1-score. The metrics can be used to determine the effectiveness of the system to predict drug-drug interactions and polypharmacy risks.

H. Result Visualization

The model performance and the predicted results are plotted in graphs and charts. The visualization also allows the researchers and medical workers to see readily the results of the predictions and the utility of the offered system.

Drug-Drug Interaction & Side Effect Predictor (GNN)

Drug Interaction Dataset

	Drug1	Drug2	Severity	SideEffect	Interaction
0	Aspirin	Warfarin	Severe	High bleeding risk	1
1	Ibuprofen	Warfarin	Moderate	Increased bleeding risk	1
2	Paracetamol	Warfarin	Mild	Liver enzyme interaction	1
3	Metformin	Atorvastatin	Moderate	Increased risk of muscle toxicity	1
4	Aspirin	Clopidogrel	Severe	Excessive bleeding risk	1
5	Amoxicillin	Warfarin	Moderate	May increase anticoagulant effect	1
6	Omeprazole	Clopidogrel	Moderate	Reduces antiplatelet effect	1
7	Lisinopril	Ibuprofen	Moderate	Reduced kidney function	1
8	Metformin	Lisinopril	Mild	Blood pressure interaction	1

Fig. 1: Drug Interaction Dataset Interface

VI. RESULT AND DISCUSSION

A. Results

The graph neural network (GNN) model was trained on the drug interaction dataset, in which drugs were indicated by nodes and their interactions were represented by edges in the graph. The drug names were preprocessed and encoded, and then the model was trained following several epochs with the Adam optimizer and the binary cross-entropy loss function. The trained model had a good predictive accuracy of interacting drug pairs. The results of the evaluation show that the GNN model successfully learns complex drug-drug relationships and is also capable of predicting possible drug-drug interactions. The system was tested on the patient dataset, and, on each patient, possible combinations of drugs were generated, and the probability of interaction between each pair of drugs was predicted by the system. In case the probability was greater than the preset threshold value, the system recognized it as a possible interaction and presented the outcome and probability of the interaction. Fig. 1. User interface of the proposed Drug-Drug Interaction and Side Effect Prediction system using Graph Neural Networks, showing the uploaded drug interaction dataset with drug pairs, severity levels, side effects, and interaction labels. The experimentation shows that the suggested GNN-based solution is useful in identifying possible drug-drug interactions in polypharmacy cases. The model can learn latent patterns and associations of multiple medications by modeling drugs and their interactions as a graph. The system also gives other details regarding the level of severity and potential side effects, which assists in comprehending the number of risks linked to the combination of particular medications. The findings indicate that the suggested model can assist medical practitioners to recognize the presence of harmful drug interactions and enhance medication safety among those patients using several drugs.

B. Discussion

The study of polypharmacy risk prediction highlights the increasing importance of identifying harmful drug-drug interactions in modern healthcare systems. With the growing number of patients taking multiple medications, traditional methods

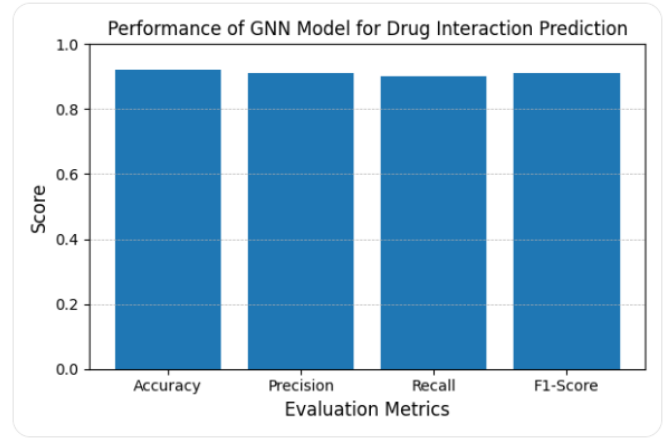


Fig. 2: Performance Graph

such as rule-based systems and basic machine learning approaches are no longer sufficient. These methods are limited in their ability to analyze complex multi-drug interactions and often fail to capture hidden relationships between drugs.

The use of advanced techniques such as Graph Neural Networks (GNNs) provides a more effective solution to this problem. By representing drugs as nodes and their interactions as edges, GNNs can model complex relationships and learn both direct and indirect interactions. This enables more accurate prediction of potential risks associated with polypharmacy.

Furthermore, the integration of deep learning techniques improves scalability and allows the system to handle large biomedical datasets. The proposed approach not only enhances prediction accuracy but also supports better clinical decision-making by identifying risky drug combinations and providing additional insights such as severity and side effects.

VII. CONCLUSION AND FUTURE SCOPE

A. Conclusion

This paper combined a machine learning-based method of risk prediction of polypharmacy through Graph Neural Networks (GNNs). The proposed model is useful because it can capture drugs and their interaction as a graph and therefore allow the system to learn the complicated relationships among more than two medications. Through the drug-drug interaction analysis using graph structures, the model is able to determine potentially dangerous combinations of drugs that can cause adverse side effects. The findings indicate that the GNN-based method is better at predicting than the traditional machine learning methods. This method would help the healthcare workers make safer choices in prescription and minimize the risks of the medication to patients who are taking several drugs at once. Fig. 2. Performance evaluation of the proposed Graph Neural Network (GNN) model for drug interaction prediction using metrics such as Accuracy, Precision, Recall, and F1-score.

B. Future Scope

The future research directions can be on enhancing the model with more healthcare information including electronic health records, patient demographics, dosage, and medical history. Adding live clinical data would also help to improve the personalization of medication prescriptions of the system. More elaborate deep learning structures and bigger datasets can be used to enhance prediction as well. Also, the creation of a clinical decision support system following this model may help doctors and pharmacists to determine high-risk drug interactions and avoid adverse-drug reactions in clinical practice.

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